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APPLICATION

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APPLICANT

National ID	19730427231120000127
Name	PATRICK MOKIWA KIVANDA
Gender	Male
Date of birth	27/04/1973
Nationality	Tanzanian
E-mail Address	pkivanda@yahoo.com
Mobile Phone Number	+255784272856

TECHNICAL DESCRIPTION

Patent type	National Patent
Title	SOLAR HEATED AIR SYSTEM
Abstract	A solar heated air system, is provided in this invention. This invention comprises: a heat harvesting system to collect the heat energy from solar; heat absorbing system to absorb heat collected from the heat collector; heat transferring system for transferring the absorbed heat to the heat buffering system; heat buffering system for balancing heat on clouding times during the day; heat exchange system to allow utilization of the stored heat by heating up compressed air in the heat storage tanks; heat storage system for storing excess heat for later use when in need especially during the night or in on heavy clouded or rainy day; and air supply system. The system utilizes appropriate fluid for transferring heat between the different systems and dry heated air for carrying heat energy for domestic and industrial utilization for various purposes.
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CLAIMS

CLAIM 1

Claim number	1
Claim description	A solar heated air system, comprising: a heat harvesting system to collect the heat energy from solar; a heat absorbing system to absorb heat collected from the heat collector; a heat transferring system for transferring the absorbed heat to the heat buffering system; a heat buffering system for balancing heat on clouding times during the day; a heat exchange system to allow utilization of the stored heat by heating up compressed air in the heat storage tanks; a heat storage system for storing excess heat for later use when in need especially during the night or in on heavy clouded or rainy day; and an air supply system.

CLAIM 2

Claim number	2

Claim description	The solar heated air system, as claimed in claim 1, whereby the heat harvesting system, comprises: a parabolic concentrator (any type of solar concentrator can work) for collecting heat energy from the sun rays; a highly heat reflector film at the concave side; a variable angle tilting structure for holding the parabolic concentrator and allowing the parapolice concentrator to follow the sun rays; and a ground structure for mounting the tilting structure.

CLAIM 3

Claim number	3
Claim description	The solar heated air system, as claimed in claim 1, whereby the heat absorbing system comprises: a big diameter pipe and a small diameter pipe held by plates separating them, arranged in such a way that the plates do not create rooms within the pipes; a heat absorbing material covers up outside of both pipes; a pipe back cover plate with a hole not more than the outside diameter of the small pipe to cover back side of the pipes; an air valve to be placed in pipes back plate for air inside the small diameter pipe when fluid is put in the system, preferably low vapour pressure fluid; a pipes front cover plate with a diameter not bigger than the outside diameter of small pipe to cover the front side of the pipes; a heat insulated coupling attached to the front end of the pipes which will be used to couple the heat absorbing system and the heat transfer system; and a vacuum valve attached to the outside big pipe to be used to remove air between the big pipe and small pipe chamber so as to create and maintain vacuum inside the pipes.

CLAIM 4

Claim number	4
Claim description	The solar heated air system, as claimed in claim 1, whereby heat transfer system between heat absorber system and heat buffering system comprises: a low vapour pressure fluid, preferably low vapour pressure fluid mixed with salty material for transferring the harvested heat from the heat absorber to the heat buffering system; a heat insulated pipe for holding low vapour pressure fluid, preferably low vapour pressure fluid mixed with salty and transferring heat between heat absorber and heat buffering system; and insulated couplings at both ends of the system for connecting the heat absorber system and the heat buffering system.

CLAIM 5

Claim number	5
Claim description	The solar heated air system, as claimed in claim 1, whereby heat buffering system comprises: a housing keeping the interconnected tanks surrounded by heat insulation materials; an interconnected tank filled with fluid, preferably low vapour pressure fluid for keeping transferred heat constant, this could be one up to several; a temperature sensor is fixed in the interconnected tank(s); a pressure sensor is fixed in the interconnected tank(s); a buffer pipe connected between heat transfer system, the interconnected tanks and heat exchanger system for transferring heat between the systems and components; and couplings fixed at ends of the buffer pipe to connect the heat transfer system, the tanks and the heat exchanger system.

CLAIM 6

Claim number 6

Claim description The solar heated air system, as claimed in claim 1, whereby heat transfer system between heat buffering system and heat exchanger system comprises:
a fluid, preferably low vapour pressure fluid, mixed with salty material or without for transferring the absorbed heat between the heat buffering system and the heat exchanger;
a heat insulated pipe for holding fluid, preferably low vapour pressure fluid, mixed with salty material or without or without and transfering heat between the heat buffering system and the heat exchanger; and
insulated couplings at both ends of the system for connecting the heat buffering system and the heat exchanger system.

CLAIM 7

Claim number 7

Claim description The solar heated air system, as claimed in claim 1, whereby the heat exchanger comprising:
an insulated heat exchanger housing for enclosing heat exchanger components;
a cylinder(s) for storing pressurized air;
heat exchanger pipe with couplings at the inlet and outlet for coupling the heat transfer system between the heat buffering system and the heat storage system as well in the heat exchanger tanks ;
a tank(s) filled with low pressure fluid, mixed with salty material or without for immersing a cylinder for storing presurized air;
fins are fixed inside the cylinder to facilitating het exchange;
a tempreture sensor is fixed in each cylinder;
a pressure sensor is fixed in each cylinder;
inlet and outlet air pipes fixed at the sides of the heat exchanger housing to the and out of the pressurized air cylinders;
a non-return inlet air pressure control value is attached to inlet pipe of the heat exchanger system before pressurized air enters the system;
a non-return outlet air pressure control value is attached to outlet pipe of the heat exchanger system before pressurized air leaves the system;
inlet and outlet air pipes for connecting the cylinders within the heat exchanger system; and
interconnected pipes for condensate collection at the bottom of the cylinder; and
a get value to let the condensate out of the system.

CLAIM 8

Claim number 8

Claim description The solar heated air system, as claimed in claim 1, whereby heat transfer system between heat exchanger system and heat storage system comprises:
a fluid, preferably low vapour pressure fluid, mixed with salty material or without for transferring the absorbed heat between the heat the heat exchanger system and the heat storage system;
a heat insulated pipe for holding fluid, preferably low vapour pressure fluid, mixed with salty material or without and transferring heat between the heat exchanger system and the heat storage system; and
insulated couplings at both ends of the system for connecting the heat exchanger system and the heat storage system.

CLAIM 9

Claim number 9

Claim description	The solar heated air system, as claimed in claim 1, whereby a heat storage system comprises: an insulated heat storage housing for enclosing heat storage components; a tank(s) filled with pieces of irregular shape objects capable of absorbing and storing heat which is then filled with low pressure fluid, mixed with salty material or without; an insulated heat transfer pipe with couplings at the inlet and outlet for coupling between the heat exchanger system and the heat storage system; an insulated inlet pipe with coupling at one end and other end connected at the bottom of tank(s), the other end is connected to the insulated heat transfer pipe installed between the heat exchanger system and the heat storage system; a temperature sensor is fixed in each tank; a pressure sensor is fixed in each tank; an outlet air pipes fixed at the top of tank(s) for transferring fluid, preferably low vapour pressure fluid, mixed with salty material or without to other heat storage units as per heat energy requirements and used for filling in the fluid in the system.

CLAIM 10

Claim number	10
Claim description	The Solar heated air system, as claimed in claim 1, and 8, whereby the air supply system comprising: a powered air compressor by any energy source preferably solar energy capable of supplying enough air to the heat exchange system as per requirements; and air pipes connecting the compressor and a hybrid air filter and drier to ensure that clean and dry air goes into the heat exchanger system.

OWNERS

OWNER 1

National ID	19730427231120000127
Name	PATRICK MOKIWA KIVANDA
Gender	Male
Date of birth	27/04/1973
Nationality	Tanzanian
E-mail Address	pkivanda@yahoo.com
Mobile Phone Number	0784272856
Residential address	Tanzania, Region Arusha, District Arusha, Ward Ilboru, Postal code 23227, Street Kaloleni, Road TAKUKURU RD, Plot number 24, Block number A 202, House number 3

REPRESENTATIVES

REPRESENTATIVE 1

PRIORITIES

PRIORITY 1

PATRICK MOKIWA KIVANDA

21/11/2024

Signature and date